Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input

Fast cytosol removes Wave-pinning's size-dependent boundary gain; an isolated-front prediction gives its absolute native-ring gains within 6.9–8.0% without fitting them. Native gain declines less in Holmes and Spring. Goryachev's gain falls strongly, but no conduction-law comparison is made for that model.

A, Native Wave-pinning C30 active-concentration kymographs for a cue overlapping the incumbent's 10%-contrast contour (edge gap −1) or separated from it (gap 4). Time starts at cue onset; cyan bars show spatial cue support. White dashed lines mark withdrawal after 95% of the target heading change; subsequent single-domain relaxation is included. Native cue peak 0.1 and support 3 are unchanged.

B, Time to turn halfway toward a continuous cue against its edge gap, at C30, 37.8 and 47.6 (Spring resting sizes 2.5, 3.15 and 3.97). Shades denote size within each model. Open triangles with upward arrows are right-censored lower bounds; grey spans mark gaps at which heading was undefined at one or more sizes. These strong-cue turning times are distinct from weak-pulse susceptibility.

C, Susceptibility to a weak Gaussian pulse, permanent displacement per peak×duration dose, against position at C40 (Spring resting size 2). Colored lines use the native translation-mode adjoint at f=0.001 (Spring mean input 0.004); grey curves and right axes show the same stationary mean active profile. The Gaussian SD is 1.5 (Spring 0.15). Signed gain is localized to the two lateral boundaries.

D, Wave-pinning shared permanent boundary displacement against edge-pulse peak divided by reference input 0.1, at C20 and C60; duration 50. Dashed lines use the weak-input gain. Ten-percent departures are bracketed between 5–20% of reference input in Wave-pinning, 2.6–5% in Holmes, 1.3–2.6% in Spring and 2.6–20% in Goryachev (neighboring tested amplitudes, pooled size ranges). Dose-matched duration sweeps agree within 0.2%.

E, Native edge gain normalized to each model's smallest size, against relative resting size. Wave-pinning's solid curve is predicted from the isolated front: $G=(\lambda/2)/(1+C/C^*)$, $C^*=76.3$, with stationary domain fraction θ and no fitted ring gains. Holmes's dashed curve is fitted to native inverse gain, $1/G=a+bC$, $C^*=a/b\approx403$. Spring's dashed conduction curve uses $C^*\approx49$, fitted to the free-conduction penalty $G(1000,C)/G(D,C)-1$ against C/D for $D\geq1$; the dash-dot curve multiplies that fit by the measured free-conduction gain rise. This empirical correction and the apparent native crossover (about 140) are distinct from an isolated-front prediction. Goryachev has points only. Spring's actual circumference includes extension; its native points use mean input 0.004, the sweep 0.0004. Fit-definition sensitivity is reported in the readme.

F, Wave-pinning adjoint edge gain against circumference at inactive diffusivities 2.5, 5, 10 (native), 20, 40, 100 and 1000. Every point uses the same weak mean input and unchanged reaction kinetics. All 35 Wave-pinning preparations retain a polarized branch, including $D_v=2.5$ at C60 where the pilot reported domain loss. Fast cytosol largely removes the native size dependence. All quantities use model-native units; none is a calibrated migration velocity or biochemical dose.

Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input · continued

$$G = \frac{\lambda/2}{1 + C/C^*}, \qquad C^* = \frac{2D_v}{\kappa \Delta u \theta (1-\theta)} \tag{3.1}$$

Wave-pinning crossover uses the isolated front and stationary domain fraction; Holmes uses the native inverse-gain regression and Spring uses the diffusivity-penalty regression, with intrinsic free-conduction rise shown explicitly.

$$\frac{D_{s,+}}{D_{s,-}} = \left(\frac{G_+}{G_-}\right)^2, \qquad \frac{k_{\text{drift},+}}{k_{\text{drift},-}} = \left(\frac{G_+}{G_-}\right)^2 \left(\frac{C_-}{C_+}\right)^2 \tag{3.2}$$

Conditional prediction for shared input statistics, response shape and correlated edge motion. The first ratio is physical boundary drift; angular decorrelation additionally includes geometry.